



Cert: 2273572, exp 5/31/28

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JOSH D SYLVIA

SUMMARY

Accomplished Principal Software Engineer and team leader who has worked at federal government agencies that uses technical skills in software design & development, infrastructure, and project management, along with written and verbal communication to provide the best software product. Responsible for recruitment, hiring, and mentoring junior software engineers with documentation and coding standards.

EXPERIENCE

NATIONAL SECURITY AGENCY (NSA)

FORT MEADE, MD

PRINCIPAL SOFTWARE ENGINEER / LEAD TECH DIRECTOR, 6/2019-PRESENT



- Managed a team of about 40 personnel consisting of Software Engineers, System Administrators, Project Managers, Customer Service, and Auditors.
- Managed all server migrations in real-time with minimal downtime. Reviewed Plans, specifications, and cost estimates before and during migration.
- Supervised all Tier I, Tier II, and Tier III JIRA support tickets.
- Directed a team to develop a security system dashboard that helps detect intrusion detection, network usage, CPU usage, memory usage, and much more.
- Responsible for recruitment, interviews, as well as mentoring junior software engineers and new hires about systems architecture, coding standards, documentation and data flows.
- Spearheaded Documentation of complex code, data flows, architecture, and network diagrams to have a clear picture of the overall set of systems.
- Organized code reviews with product team to get feedback for Software Engineers to Develop.
- Using Selenium, built a unit testing system for a web application that reduced bugs by 25%.
- Used Playwright and AWS, built an application to access the web using a variety of browsers.
- Adopted COTS software and 3rd party software such as; AI / ML video analytic and Deepfake technologies into existing software application.
- Working relationships with outside partners to build state-of-the-art techniques to receive and analyze data.
- Built a workflow management system that sets part of the system to working, degraded, broken, or archive. Workflow management system would remove broken or unusable workflows from customer view and add documentation on how to fix the workflow.
- Designed a UI to be more accessible for disabled users using JAWS screen reader.
- Overseen updates to software vulnerabilities and End Of Life (EOL) products as well as server migrations to get off older hardware.

BARRACUDA NETWORKS

SAN JOSE, CA

SOFTWARE ENGINEER - DEVOPS PLATFORM TEAM, 1/2018-5/2019



- Worked on a Linux development machine to build virtual machines for ESXi, HyperVisor, XenCenter, AWS, Google Cloud, and Azure.
- Installed HyperVisor, VMWare, and XenCenter tools for on premise Virtual Machines.
- Created custom scripts that would install applications and services on different platforms.
- Modified iptables pre-routing rules to point to a specific destination port that contained the End User License Agreement (EULA).
- Assisted in building a complete Software As A Service (SaaS) Development Structure.
- Helped in designing a complete database structure for SaaS project.
- Built a Docker web server container that uses python flask.
- Designed functions that would be able to communicate with Azure, AWS, and GCP.
- Maintained MySQL and Postgres databases.
- Created RESTFUL API's that would accept JSON post request and bill the customers according to their billing model.



- Supported logging that gave details of transactions processed and used sendmail daemon to send mail alert when logging hit a critical error.

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION GREENBELT, MD

WEB AND APPLICATION DEVELOPMENT INTERN, 6/2017-8/2017

- Wrote Perl scripts and used keywords to locate files in a hierarchical file system.
- Used Perl scripts along with string manipulation to redesign and rebuild a new file system.
- Update MySQL tables and fields.
- Tested a Docker container in both Windows and Linux environment.

SMART SOURCE, INC.

ST CHARLES, IL

PART-TIME IT CONSULTANT, 6/2008-6/2017

- Consultant for remote operations.
- Installed and configured network devices.
- Removed virus and malware from infected systems.
- General computer maintenance along with replacement and installation of new hardware.

XCITE NETWORKS

SCOTTS VALLEY, CA

FIELD IT TECHNICIAN, 10/2003-6/2008

- Provided customers with Desktop Support, System and Network Administration.
- Designed VOIP network infrastructure for Small to Large businesses.
- Configured ports in various routers for VOIP traffic using QOS.
- Assisted customers with tool selection of tools for network management systems.

EDUCATION

SAN JOSE STATE UNIVERSITY (SJSU), 3.33 GPA

SANTA CLARA, CA

B.S. SOFTWARE ENGINEERING, 12/2017

- Member of Computer / Software Engineering (CSE) club.

CABRILLO COLLEGE, 3.25 GPA

APTOS, CA

A.S. COMPUTER SCIENCE, 5/2015

- Treasurer of Student Senate and Engineering Abroad.

CERTIFICATE

Certified Information Systems Security Professional (CISSP)

- Certificate Number: 2273572, Certificate Cycle:6/1/25-5/31/28

Skills Certification: LINUX SYSTEM ADMINISTRATION, 5/2015

Skills Certification: MICROSOFT SYSTEMS ADMINISTRATION, 5/2015

Skills Certification: CISCO CERTIFIED NETWORK PROFESSIONAL, 5/2015

**SKILLS &
ABILITIES**

- **Programming Languages and Frameworks:** Angular, React, C, C++, C#, Java, Kotlin, Objective-C, Perl, Python, Go, Html, CSS, Shell, JavaScript, and Swift.
 - **Operating Systems:** Linux(RedHat/Debian), Unix, Mac, Android, IOS, Win Server, & more.
 - **Databases:** SQL, MySQL, Postgres, MongoDB.
 - **DevOps, Cloud, Tools and programs:** AWS, Azure, GCP, EC3, S3, NiFi, Kafka, Ansible, Terraform, SDLC, Scrum, VPNs, Docker, Kubernetes, Elasticsearch, microservices, Git, JIRA, Confluence, Agile, Scrum, CI/CD, DevSecOps, IntelliJ, PyCharm, StarUML, AutoCAD, PSpice, Xcode, Android Studio, Eclipse, NetBeans, Splunk, SwiftUI, Visio, WireShark, Jupyter Notebook and MS Office.
 - **Computer Networking:** OSI model, Network security communication, OSPF, IGRP, EIGRP, BGP, VLAN, VRRP, InterVlan Routing, STP, Ipsec tunneling, Load Balancers, VRRP.
 - **Generative AI:** Windsurf, Claude, Copilot, ChatGPT, Gemini.
 - **Testing:** UI Testing, API Testing, White box testing, Black Box Testing, Unit Testing, Test Automation, SAST, DAST, User Acceptance Testing (UAT), Integration Testing, Regression Testing.
 - **Cloud Architecture:** IaaS, PaaS, DaaS, SaaS, FaaS, SaaS, VPC, SASE.
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- **Security and Security Controls:** Role-Based Access Control (RBAC), DAC, Gap Analysis, CVEs, Threat Modeling, Cyber Threat Intelligence, Metrics, KPIs, KPI Metrics, KPI Analysis, Log Data Analysis, API Security, Security as Code, Cloud Security, Security Testing, SOC, SOC2, FedRAMP, Web Application Security, RMF process, Incident Response, IAM ,Risk Management, PII, PHI, EDR, DLP, MDR, Change Management, Configuration Management, HIPAA, GDPR, CCPA, SLA, Root Cause Analysis, SIEM, SOAR.
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LEADERSHIP

- Principal Software Engineer and Lead Tech Director of about 40 personnel.
 - Speaker at major online Skype event that spanned five different countries.
 - Treasurer of Student Senate Cabrillo College.
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